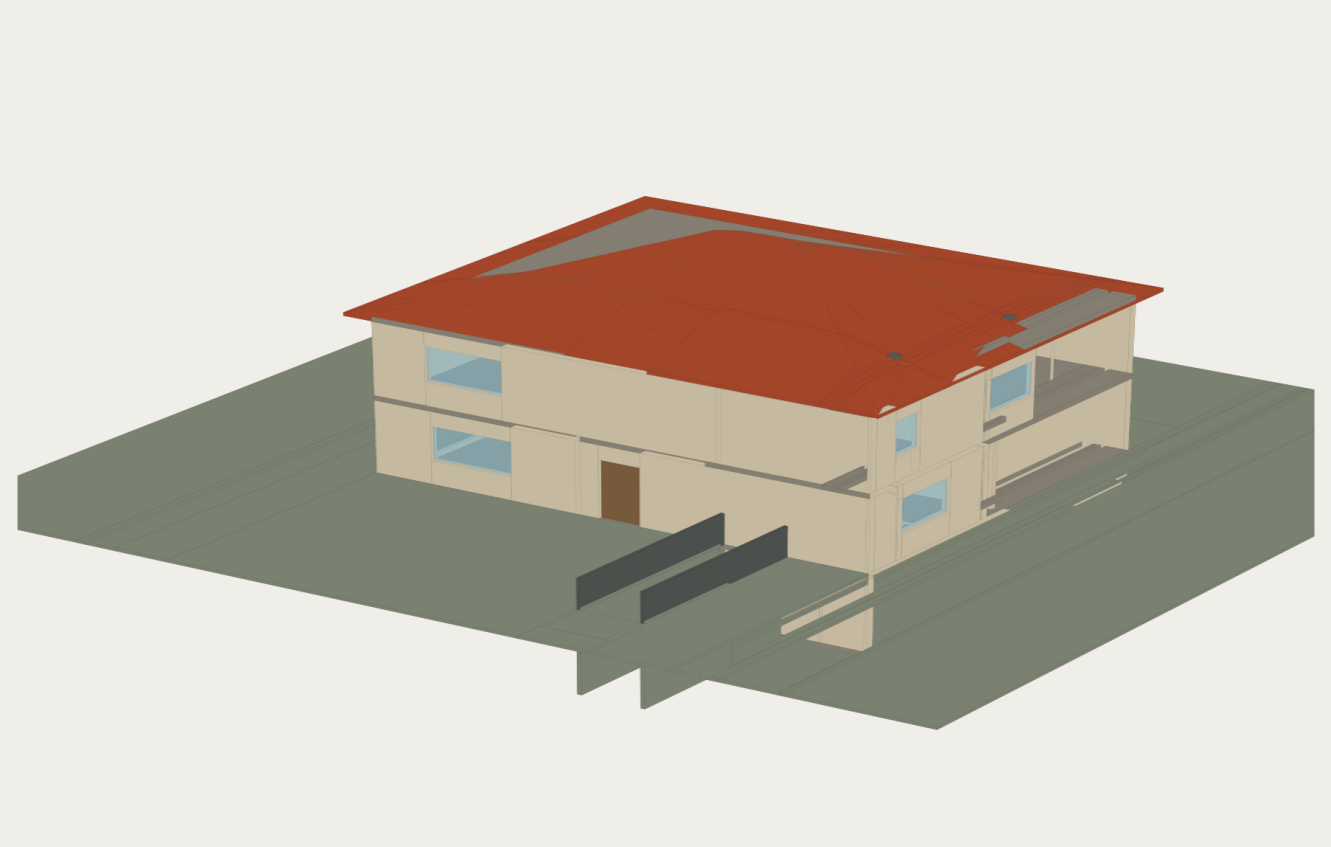
JOHANSSON HOUSE Final technical and validation report

REVISION F | 13 OCTOBER 2026 | Continuing the Revision E game-development project

Acceptance: PASS within the coordinated game-production reference scope. All 62 independent checks and 31 reference simulation tests passed. No unresolved defects remain within those tested boundaries. The package supplies drawings, model geometry, service networks and an implementation contract; it does not assert real-world constructability or a tested deployment in Johansson World.



The house retains the 12 x 10 m footprint and its original 19 room IDs. The programme combines Johansson's workshop and club room, Thuan's sewing studio, domestic rooms, sauna/shower and plant space. The inherited house brief dates its setting to circa 1995; it can be used as an existing house in the wider town setting.

Deliverable control

Item	Final content
Architecture	21 editable DXFs; synchronized floor plans, roof, four elevations, sections, site and functional / physical services.
3D assets	Architecture: 708 nodes, 9096 triangles. Services: 285 nodes, 9120 triangles.
Systems	84 coordinate routes, 19 luminaires, 11 socket points, 125 reserved penetrations and two roof sleeves.
Evidence	Machine-readable checks, executable regression suite, source inputs, migration map and final defect register.

1. Coordinate and geometry control

Control	Value / rule
CAD and JSON	Millimetres; X east, Y north, Z up. Origin at southwest exterior ground corner.
GLB	Metres, Y up: (X,Z,-Y)/1000. Scale 1 on import.
Finished floors	B1 -2800; G0 0; L1 +2800 mm. Slab top is finished floor; the former 100 mm offset is removed.
Wall and slab	Exterior 200; internal 120; slab 180 mm nominal. Game representations only.
Roof	Structural deck 5420-5600; tile underside 5680; eaves 5800; ridge 8000; 400 overhang.
Stair	16 x175 risers; 7 x250 treads per flight; 900 flight/landing; 200 centre gap. Hole 2000 x2650.
Clear side passage	1100 mm after guarding. 500 mm reference capsule.
Basement exterior	1000 wide garden stair, 16 x175 risers, 250 treads; 1400 x1300 bottom landing and separate storm sump.

The inherited room rectangles were reconciled with partitions. The central passage was widened by moving the east partition to X=8300. Right-side rooms consequently have a 3440 mm clear nominal width; they retain their names and IDs. Storage and linen occupy two joined clear zones. Scheduled area is clear planning-zone area, including the stair zone where applicable; it is not a certified statutory floor-area measurement.

Assemblies and appearance

Exterior: 200 mm nominal reinforced concrete; game wall finish weathered cream; exterior waterproof membrane below grade

Partitions: 120 mm nominal concrete/masonry visual walls; isolated sliding apertures

Floors: 180 mm nominal slab; top equals FFL; additional floor finish represented in material, not elevation offset

Foundation: 400 mm nominal perimeter footing, 300 mm deep; geometric game representation only

Roof: hipped tile shell; underside 5680; eaves5800/ridge8000; 400 overhang; lower deck5420-5600

Wet_zones: tiled finish, floor drain/trap state, watertight sleeve and equipment isolation

Shutters: optional weather-state art variant, no structural claim

Weathering: salt staining at fittings, roof tile mottling, worn timber sliding doors

Roof_sleeves: 140 x140 shaft apertures around100 mm vent/flue; 20 radial clearance; four-sided weather collars provided as cosmetic roof solids

2. Room schedule

ID / level	Room	Clear zone coordinates [x0,y0,x1,y1] mm	Zone area m2
B1-ROOM-01	Electrical workshop	[200, 200, 4840, 4940]	21.994
B1-ROOM-02	Club reading room	[200, 5060, 4840, 9800]	21.994
B1-ROOM-03	Stair hall	[4960, 200, 8240, 6440]	20.467
B1-ROOM-04	Boiler and pumps	[8360, 200, 11800, 3440]	11.146
B1-ROOM-05	Sauna and shower	[8360, 3560, 11800, 7040]	11.971
B1-ROOM-06	Storage	[4960, 6560, 8240, 9800]; [8360, 7160, 11800, 9800]	19.709
G0-ROOM-01	Living room	[200, 200, 4840, 4940]	21.994
G0-ROOM-02	Kitchen and dining	[200, 5060, 4840, 9800]	21.994
G0-ROOM-03	Stair hall	[4960, 3560, 8240, 9800]	20.467
G0-ROOM-04	Thuan sewing studio	[8360, 200, 11800, 4940]	16.306
G0-ROOM-05	Entry and genkan	[4960, 200, 8240, 3440]	10.627
G0-ROOM-06	WC and utility	[8360, 5060, 11800, 7040]	6.811
G0-ROOM-07	Covered terrace access	[8360, 7160, 11800, 9800]	9.082
L1-ROOM-01	Primary bedroom	[200, 200, 4840, 4940]	21.994
L1-ROOM-02	Guest bedroom	[200, 5060, 4840, 9800]	21.994
L1-ROOM-03	Stair hall	[4960, 200, 8240, 6440]	20.467
L1-ROOM-04	Bathroom	[8360, 200, 11800, 3440]	11.146
L1-ROOM-05	Study	[8360, 3560, 11800, 7040]	11.971
L1-ROOM-06	Linen and landing	[4960, 6560, 8240, 9800]; [8360, 7160, 11800, 9800]	19.709

All 19 rooms are reached by the open-door plan test with a 250 mm capsule radius. Sliding doors avoid swing conflicts in the compact circulation zone. Window W-B1 is daylight through a lightwell, not a designated escape opening. Garden stairs and internal stairs provide the two game-access routes to the basement.

3. Opening schedule

Opening ID	Level	Side / axis position	Start-end mm	Sill-head above FFL mm	Operation
D-BEXT	B1	south	8800-9800	0-2100	sliding door
W-B1	B1	west	5900-6900	1100-2100	operable window
D-B1-WEST-S	B1	x=4900	2300-3200	0-2100	sliding door
D-B1-WEST-N	B1	x=4900	7500-8400	0-2100	sliding door
D-B1-EAST-S	B1	x=8300	2300-3200	0-2100	sliding door
D-B1-EAST-M	B1	x=8300	4500-5400	0-2100	sliding door
D-B1-EAST-N	B1	x=8300	7900-8800	0-2100	sliding door
D-B1-HALL-N	B1	y=6500	7300-8200	0-2100	sliding door
D-ENTRY	G0	south	5550-6550	0-2100	sliding door
W-LIVING	G0	south	1400-3400	850-2100	operable window
W-SEW	G0	east	1200-3000	850-2100	operable window
W-KITCHEN	G0	west	6900-8500	850-2100	operable window
D-TERRACE	G0	north	8800-9800	0-2100	sliding door
W-DINING	G0	north	1600-3400	850-2100	operable window
D-G0-WEST-S	G0	x=4900	2300-3200	0-2100	sliding door
D-G0-WEST-N	G0	x=4900	7500-8400	0-2100	sliding door
D-G0-EAST-S	G0	x=8300	2300-3200	0-2100	sliding door
D-G0-EAST-M	G0	x=8300	5800-6700	0-2100	sliding door
D-G0-EAST-N	G0	x=8300	7900-8800	0-2100	sliding door
D-G0-ENTRY-HALL	G0	y=3500	7300-8200	0-2100	sliding door
W-BED	L1	south	1300-3200	850-2100	operable window
W-BATH	L1	east	900-1800	850-2100	operable window
W-STUDY	L1	east	4400-6100	850-2100	operable window
W-GUEST	L1	west	6500-8300	850-2100	operable window
W-LANDING	L1	north	8600-10400	850-2100	operable window
D-L1-WEST-S	L1	x=4900	2300-3200	0-2100	sliding door
D-L1-WEST-N	L1	x=4900	7500-8400	0-2100	sliding door
D-L1-EAST-S	L1	x=8300	2300-3200	0-2100	sliding door
D-L1-EAST-M	L1	x=8300	4500-5400	0-2100	sliding door
D-L1-EAST-N	L1	x=8300	7900-8800	0-2100	sliding door
D-L1-HALL-N	L1	y=6500	7300-8200	0-2100	sliding door

Start/end are measured on X for north/south openings and on Y for east/west openings. Internal axis X means a wall at fixed X; start/end run along Y. Internal axis Y means a wall at fixed Y; start/end run along X. The opening schedule generates wall cut-outs and elevation apertures directly.

4. Electrical systems

The selected simulation supply is single-phase three-wire 100/200 V at 60 Hz. Meter feeds main isolation and B1/G0/L1 boards. 100 V branches are assigned to L1 or L2; 200 V branches span both legs. This is a gameplay assumption. Schedule amperes are simple W/V estimates; the game trip budgets are not fuse, breaker or cable installation ratings.

Load group	W	V	Leg	A	Game trip W
LIGHT_G	180	100	L1	1.8	225.0
SOCKET_G	1500	100	L2	15.0	1875.0
SEWING	300	100	L1	3.0	375.0
LIGHT_L	180	100	L2	1.8	225.0
AC_L	900	200	both	4.5	1125.0
WORKSHOP	1500	100	L1	15.0	1875.0
SAUNA	4500	200	both	22.5	5625.0
PUMP	400	100	L2	4.0	500.0
LIGHT_B	180	100	L2	1.8	225.0
AC_G	900	200	both	4.5	1125.0
VENT	180	100	L1	1.8	225.0
HEATER_E	250	100	L2	2.5	312.5
STORM_PUMP	350	100	L1	3.5	437.5
SOCKET_L	1000	100	L2	10	1250

Connected group load 12320 W; 0.6 diversity gives 7392.0 W against a 12000 W main game budget. The 19 room luminaires share the 180 W allowance on each level. The 11 socket points inherit the relevant group budget; do not add each point as another group load.

Room luminaire and socket coordinates are in electrical_fixtures. Route data covers every functional electrical edge, including local room drops. Main/board/branch isolation propagates to descendants. An open-conductor or leakage state survives a reset until repaired through the verified-isolation gate.

5. Potable water and hot water

Water service enters through meter, shutoff and cold manifold. The heater feeds the kitchen, first-floor bathroom and sauna shower through separate hot ports. G0 WC and laundry are cold-only. Each branch has a physical endpoint and visual envelope; trunk 32 mm and branch 20 mm values are game geometry assumptions. No real pipe selection or backflow certification is asserted.

Fixture	Flow L/s	Hot available
KITCHEN	0.12	yes
G0_WC	0.1	no
L1_BATH	0.2	yes
LAUNDRY	0.15	no
SAUNA_SHOWER	0.15	yes

Game source pressure is 300 kPa; simultaneous cold demand is 0.6 L/s with 1.0 L/s assumed source availability. The validation applies Darcy friction factor 0.025, route length/diameter and elevation rise. Every scheduled fixture retains over 100 kPa in that model. This is a reproducible model adequacy check, not a verified utility supply calculation.

Fixture	Residual model pressure kPa
KITCHEN	277.22
G0_WC	280.25
L1_BATH	251.02
LAUNDRY	279.36
SAUNA_SHOWER	293.6

Hot-water storage is 180 L at 60 C with a 40 C mixed delivery target. Availability requires cold supply, fuel, healthy heater and HEATER_E power. Reference recovery is 0.05 L/s. Relief, backflow and temperature states belong to the game appliance contract; flue is separately routed through a roof sleeve.

6. Sanitary, storm and basement water

Kitchen, WC, laundry and first-floor bath join the soil stack through descending gravity routes. Horizontal sanitary segments fall at least 2%. The stack continues as a roof vent. The B1 shower cannot drain to the higher island sewer by gravity; a dedicated lift with 1 L/s capacity and 4 m model head feeds the stack. A check-valve state prevents simulated backflow when stopped. The assumed sewer terminal invert is -1040 mm.

Storm is a separate network. The two roof downpipes each have 3 L/s game capacity. For 100 mm/h rain, runoff coefficient 1 and roof plan 138.24 m2, runoff is 3.84 L/s. Two downpipes provide 6 L/s. The courtyard drain and lightwell drain join a dedicated storm sump, rather than sanitary wastewater. Storm lift is 5 L/s at 5 m game head. Roof storm receptor and pumped courtyard outfall are separate from sewer.

Courtyard and lightwell drains are physically represented. Yard grade is assumed at 0, with real voids for the stair court and lightwell. Guarded edges prevent inadvertent access; stair portals can be closed by flood state. The sump raises an alarm above 150 L and is bounded at 200 L. The sanitary sump alarms above 40 L and is bounded at 60 L. These are game scenario capacities, not real flood-risk analysis.

Drainage invariants

All gravity paths descend; no sanitary gravity branch is routed uphill.

Pumped branches are explicitly typed and depend on their electrical supply.

Water shutoff stops new shower inflow; restored electricity cannot clear a blocked discharge.

Storm and sanitary do not share an outfall or pump state.

Continuous vent/soil centre-line is intentional; other cross-family tube collisions are zero.

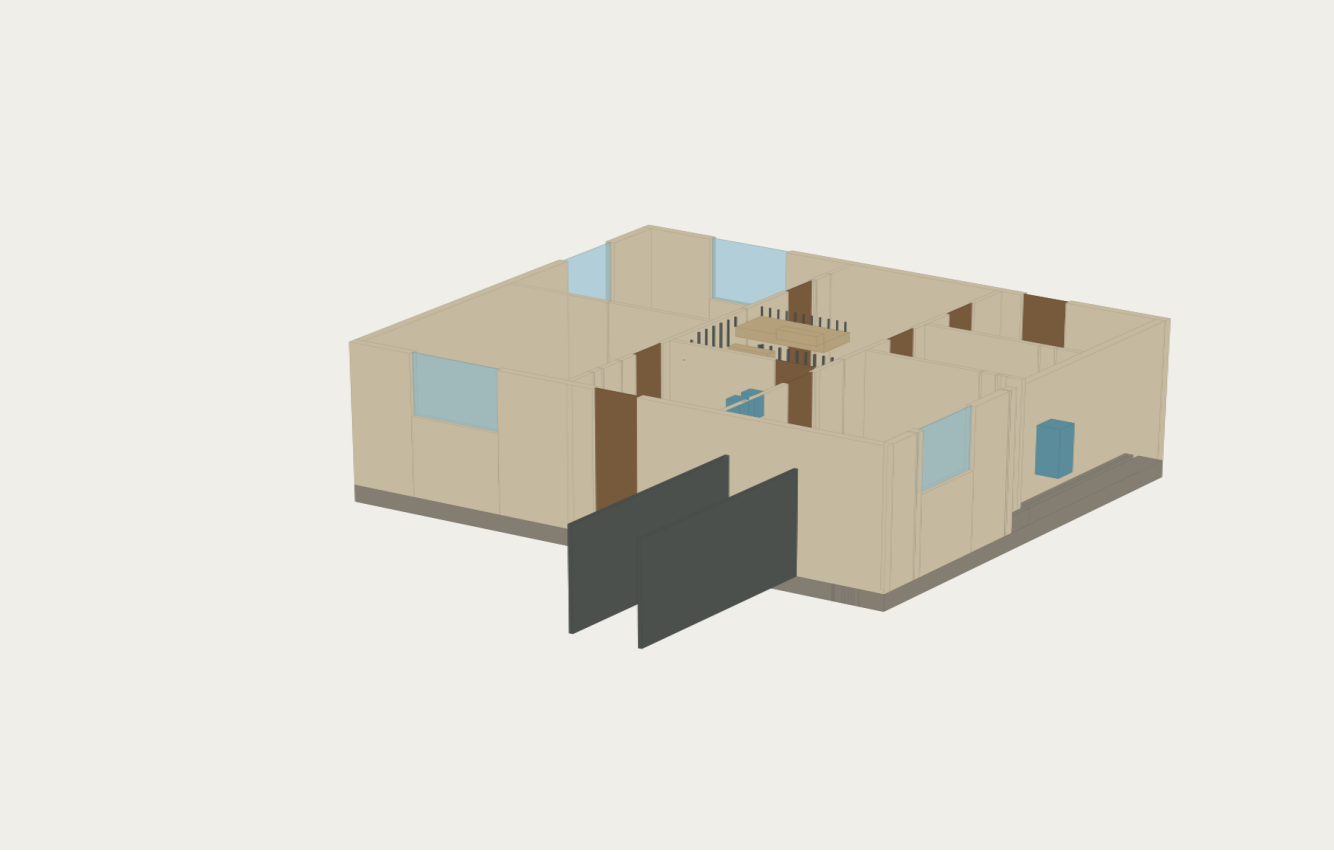
7. HVAC and ventilation

Two selected cooling zones are defined: G0_AC and L1_AC, each 4 kW for 50 m2 at an assumed 80 W/m2 game demand. Remaining rooms use operable windows and local fans. This is deliberately a room-climate game model rather than a whole-building cooling load calculation.

Fan	Target m3/h	Behaviour
B1_FRESH	400	Filtered make-up air
B1_EXHAUST	180	General extract; own route
SAUNA_EXHAUST	100	Independent wet extract
SMOKING_EXHAUST	120	Independent smoke extract; negative pressure state

The full target balances supply 400 against exhaust 180+100+120. Each fan can be switched or failed independently. All fans depend on VENT power. The smoking room does not exhaust through the sauna or general extract circuit. Air-conditioning refrigerant and condensate use separate ports; condensate routes descend to an air-gap storm receptor. A condensate blockage disables the affected cooling unit in the reference model.

Maintenance envelopes use a 600 mm clearance target and approach markers. These are metadata for host navigation projection; their cosmetic equipment meshes are non-colliding. Sealed hot-water heater flue is independently routed through the roof, rather than connected to an extract duct.



8. Game asset and persistence specification

Import the two GLBs at scale 1. Apply (X,Z,-Y)/1000 only to CAD/JSON points.

Use stable node IDs, not mesh indices. Original room IDs survive; the complete E migration map has explicit preserve/alias/retire actions.

Use collision.json solids and ramp/landing proxies. Glass, services, fixtures and equipment envelopes are non-static; doors require dynamic panels.

Reference capsule: radius250 mm, height1800 mm, step200 mm and slope40 degrees. Guarded side passage is1100 mm.

Persist breaker, valve, fan, door, room light, fault, verified-isolation, fuel/heater and volume states under schema2. Reject unknown schemas/IDs and out-of-range volumes.

Group levels, roof, site and services for indoor camera suppression, static batching and room occlusion. Reference asset budgets are satisfied, but FPS belongs to the host runtime.

Flood alarm gates basement links; verified isolation gates repair; clearing a breaker trip never clears its underlying fault.

Game_Integration.md gives the ordered import, behaviour and live-host acceptance contract. source/reference_sim.py is the tested state implementation. The delivery does not modify a repository or claim successful live-engine import, camera behaviour or frame rate.

Reproduction

Install source/requirements.txt. Run source/build.py, source/test_sim.py and source/validate.py in that order. The builder regenerates native assets, CAD, route and room schedules, and the drawing book. The final report, acceptance decision and defect register are issue-control documents. Source Revision E JSON is provenance only; its old holds are superseded by this report.

9. Defect resolution register

Defect	Disposition	Evidence / resolution
Room-zone / partition overlap	CLOSED	19 room IDs preserved; exact floor zones exclude partitions. Doors checked with a 500 mm capsule and room flood fill.
Duplicate placeholder roof	CLOSED	One manifold hipped roof; lower deck is a separate ceiling slab. Ridge 8000, eaves 5800; roof has flue and vent apertures.
Stair run and landing	CLOSED	16 risers at 175; 7 treads at 250 per flight; 900 landing and flight width. Opening 2000 x 2650; 1100 clear guarded side passage.
Basement access / flood / daylight	CLOSED FOR GAME SCOPE	1000 wide garden stair; 1400 x 1300 bottom court; drained west lightwell. Dedicated storm lift, flood alarm and entry gating.
Engineering load and installation sizing	SCOPE EXCLUSION	Game electrical budgets, pressure/fall calculations, pipe/duct envelopes and HVAC assumptions are defined. No construction certification or installation ratings are claimed.
Elevation / section consistency	CLOSED	Openings, floor levels, roof profiles and cut geometry are generated from the same manifest.
Assemblies / product selection	CLOSED FOR GAME SCOPE	Concrete wall/slab, foundation, tile roof, sealed sleeves, wet-zone and weathering specifications established; manufacturer procurement is outside game scope.
Preliminary MEP routes	CLOSED	84 spatial routes, declared physical ports, 125 wall/slab/grade reservations, independent clearance and fall checks.
Roof overlay audit	CLOSED	Inherited GLBs are superseded; one named ROOF-HIPPED mesh exported. No visual flat roof overlay retained.
Stair headroom / guarding	CLOSED	28 tread-centre headroom samples meet 2000 mm; sloping handrails, opening guards, exterior guards and ramp collision proxies provided.
Closed room / solid geometry	CLOSED	Room wall-overlap and access checks pass. Box solids closed individually; roof edge valence is two. Apertures remain intentionally open.
Blocked hall portals discovered in final audit	CLOSED	Entry and north hall sliding apertures shifted into side passage; all room zones now accessible.
Cross-family service clashes discovered in final audit	CLOSED	Separated route lanes and equipment ports; final distinct-family collision count zero. Same-family shared routes and continuous vented soil stack are explicit.
Game runtime acceptance	SCOPE EXCLUSION	Tested reference simulation and integration contract delivered. Live repository performance, animation, camera and engine imports require host acceptance; no live integration claim is made.

10. Independent validation record

The validator checks exported data, not merely the exporter return status. The 62 checks cover DXF audits, mesh indices/areas, route-coordinate parity, capsule access, room partition conflicts, endpoint ports, gravity falls, envelope clearance, roof closure, stair headroom, electrical demand, game water pressure, storm capacity and balanced ventilation. The 31 regression cases independently verify the reference state behaviour.

Check	Result	Evidence summary
unique stable IDs	PASS	{'count': 796}
all positive geometry volumes	PASS	No reversed or zero-size solid
service versus architecture solids	PASS	{'clashes': [], 'total': 0}
gravity routes descend with specified fall	PASS	[]
spatial routes equal functional endpoints	PASS	[]
all door apertures pass 500 mm capsule width	PASS	[]
all room zones reachable by 500 mm capsule on open-door plan	PASS	{'B1-ROOM-01': 1816, 'B1-ROOM-02': 1816, 'B1-ROOM-03': 1097, 'B1-ROOM-04': 862, 'B1-ROOM-05': 910, 'B1-ROOM-06': 1484, 'G0-ROOM-01': 1816, 'G0-ROOM-02': 1816, 'G0-ROOM-03': 1076, 'G0-ROOM-04': 1300, 'G0-ROOM-05': 824, 'G0-ROOM-06': 460, 'G0-ROOM-07': 682, 'L1-ROOM-01': 1816, 'L1-ROOM-02': 1816, 'L1-
stair vertical headroom >=2000 mm	PASS	{'minimum_mm': 2625, 'samples': 28}
stair rise run and available enclosure	PASS	{'rise': 2800, 'run': 2650, 'width': 2000, 'side_passage': 1140}
Johansson_House_Architecture.glb valid binary meshes	PASS	[]
Johansson_House_Architecture.glb performance budget	PASS	{'nodes': 708}
Johansson_House_Services.glb valid binary meshes	PASS	[]
Johansson_House_Services.glb performance budget	PASS	{'nodes': 285}
A_B1_Coordinated.dxf CAD audit	PASS	{'entities': 109, 'errors': 0, 'fixes': 0}
A_G0_Coordinated.dxf CAD audit	PASS	{'entities': 121, 'errors': 0, 'fixes': 0}
A_L1_Coordinated.dxf CAD audit	PASS	{'entities': 103, 'errors': 0, 'fixes': 0}
A104_Roof.dxf CAD audit	PASS	{'entities': 11, 'errors': 0, 'fixes': 0}
A201_Elevations.dxf CAD audit	PASS	{'entities': 251, 'errors': 0, 'fixes': 0}
A301_Sections.dxf CAD audit	PASS	{'entities': 113, 'errors': 0, 'fixes': 0}
C101_Site_Utilities.dxf CAD audit	PASS	{'entities': 27, 'errors': 0, 'fixes': 0}
R_condensate_3D.dxf CAD audit	PASS	{'entities': 6, 'errors': 0, 'fixes': 0}
R_electrical_3D.dxf CAD audit	PASS	{'entities': 100, 'errors': 0, 'fixes': 0}
R_flue_3D.dxf CAD audit	PASS	{'entities': 4, 'errors': 0, 'fixes': 0}
R_hvac_3D.dxf CAD audit	PASS	{'entities': 10, 'errors': 0, 'fixes': 0}
R_potable_cold_3D.dxf CAD audit	PASS	{'entities': 20, 'errors': 0, 'fixes': 0}
R_potable_hot_3D.dxf CAD audit	PASS	{'entities': 8, 'errors': 0, 'fixes': 0}
R_refrigerant_3D.dxf CAD audit	PASS	{'entities': 6, 'errors': 0, 'fixes': 0}
R_sanitary_3D.dxf CAD audit	PASS	{'entities': 18, 'errors': 0, 'fixes': 0}
R_storm_3D.dxf CAD audit	PASS	{'entities': 12, 'errors': 0, 'fixes': 0}
R_vent_3D.dxf CAD audit	PASS	{'entities': 4, 'errors': 0, 'fixes': 0}
E101_One_Line.dxf CAD audit	PASS	{'entities': 151, 'errors': 0, 'fixes': 0}
P101_Water.dxf CAD audit	PASS	{'entities': 34, 'errors': 0, 'fixes': 0}
P102_Drainage.dxf CAD audit	PASS	{'entities': 46, 'errors': 0, 'fixes': 0}
M101_HVAC.dxf CAD audit	PASS	{'entities': 38, 'errors': 0, 'fixes': 0}
electrical route DXF/manifest parity	PASS	{'polylines': 49}
vent route DXF/manifest parity	PASS	{'polylines': 1}
condensate route DXF/manifest parity	PASS	{'polylines': 2}
hvac route DXF/manifest parity	PASS	{'polylines': 4}

Check	Result	Evidence summary
sanitary route DXF/manifest parity	PASS	{'polylines': 8}
potable_hot route DXF/manifest parity	PASS	{'polylines': 3}
refrigerant route DXF/manifest parity	PASS	{'polylines': 2}
storm route DXF/manifest parity	PASS	{'polylines': 5}
potable_cold route DXF/manifest parity	PASS	{'polylines': 9}
flue route DXF/manifest parity	PASS	{'polylines': 1}
every Revision E asset has a migration action	PASS	{'migrated': 58}
migration aliases resolve	PASS	Retired dependencies explicitly documented
HVAC supply/extract design balance	PASS	{'B1_FRESH': 400, 'B1_EXHAUST': 180, 'SAUNA_EXHAUST': 100, 'SMOKING_EXHAUST': 120}
roof stormwater design capacity	PASS	{'rain_mm_h': 100, 'roof_plan_m2': 138.24, 'runoff': 1, 'roof_flow_L_s': 3.84, 'downpipes': 2, 'downpipe_game_capacity_L_s_each': 3}
game electrical main demand budget	PASS	{'connected_W': 12320, 'diversified_W': 7392.0, 'budget_W': 12000}
roof closed manifold including sleeve walls	PASS	{'edges': 498, 'non_manifold_edges': 0}
roof has vent and flue apertures with 20 mm radial clearance	PASS	[[11330, 5930, 11470, 6070], [11030, 2130, 11170, 2270]]
no inherited visual flat roof overlay	PASS	One hipped exterior roof and one lower structural ceiling deck
room zones do not overlap wall solids	PASS	[]
every named room has a controlled luminaire	PASS	{'luminaires': 19, 'fixture_total': 30}
equipment service port offsets <=250 mm	PASS	[]
route scheduled lengths equal spatial polylines	PASS	[]
electrical and potable graphs fully routed	PASS	[]
distinct service families have no tube envelope collisions	PASS	{'clashes': [], 'intentional_shared': 'sanitary and vent occupy one continuous vented stack; same-family routes may share network trunks or conduit runs'}
game cold-water pressure exceeds 100 kPa at design fixtures	PASS	{'residual_kPa': {'KITCHEN': 277.22, 'G0_WC': 280.25, 'L1_BATH': 251.02, 'LAUNDRY': 279.36, 'SAUNA_SHOWER': 293.6}, 'model': 'Darcy factor 0.025; upstream 0.6 L/s; fixture demand from schedule; nominal game envelopes; not engineering pipe sizing'}
external and internal stair proxy slopes within 40 degrees	PASS	{'slope_deg': 34.99, 'end_step_mm': 175}
opening guards preserve 1100 mm physical side passage	PASS	{'clear_mm': 1100, 'capsule_width_mm': 500}
reference simulation regression suite	PASS	{'tests_run': 31, 'failures': 0, 'errors': 0, 'pass': True, 'scope': 'reference simulation only'}

Acceptance boundary

PASS: native reference geometry and scheduled systems, scoped spatial checks, editable drawings, model exports and executable reference logic. No in-scope defects remain. EXCLUDED: surveyed land data, structural engineering, real electrical/pipe/duct installation design, manufacturer procurement, regulatory approval, and live-host runtime testing. Those exclusions are not represented as passed checks.

Only this final issue is supplied. The obsolete inherited DXFs/GLBs and interim validation files are omitted from the user-facing root. Machine-readable records and checksums accompany this issue.